

Lightweight Post-training for Controllable Vietnamese Summarization

Tran Dinh Phuc

Hanoi University of Science and Technology Viettel Artificial Intelligence Center
HUST, Vietnam
tdphuc.work@gmail.com

Xuan-Dung Doan

Viettel Group, Vietnam
dungdx4@viettel.com.vn

Abstract

Controllable Vietnamese summarization requires a model to preserve source content while also obeying explicit output budgets. We present an empirical study of a lightweight post-training pipeline that combines instruction-conditioned LoRA supervised fine-tuning (SFT) with a custom group relative policy optimization (GRPO) loop for two Qwen3-4B backbones. The main paper evaluates the resulting post-trained systems on four datasets (VietNews, WikiLingua, ViMs, and VLSP) and situates them against contextual baselines such as VietAI, GPT-4o, GPT-3.5-Turbo, Qwen3-14B, and Llama3.3-70B-Instruct using ROUGE-2 and absolute summary length distance. On the two held-out single-document datasets that support our strongest claims, the final SFT + GRPO v5 systems substantially improve over their raw Qwen3 backbones and compare favorably to several historical, non-protocol-matched external rows while keeping low length distance; the Instruct system reaches 29.62 ROUGE-2 on VietNews, and the Base system reaches 26.35 on WikiLingua. We keep ViMs and VLSP visible in the benchmark tables as exploratory multi-document diagnostics because their current test pools overlap with project training or validation data, and we use targeted ablations to show that SFT is the dominant improvement stage, that warm-starting GRPO from the SFT checkpoint is better than applying GRPO directly to the pretrained backbone, and that the safeguarded reward design sharply reduces failure-prone behavior relative to the earlier GRPO setup. The code for the project is available at https://github.com/tranphuc15122004/PoML_for_summary.

1 Introduction

Controllable summarization is useful when systems must meet strict output budgets, such as headline generation, short mobile previews, or

template-bound reporting. In these settings, a model must preserve salient content while also satisfying an explicit length target. That combination is still difficult for lightweight open-weight language models, especially in Vietnamese, where controllable summarization resources and post-training analyses remain limited.

This paper studies a practical post-training recipe for that setting as an empirical case study of lightweight Vietnamese controllable summarization. Our focus is not to enumerate model variants for their own sake, but to explain a method (instruction-conditioned LoRA-SFT followed by custom GRPO) and to position its final outputs against external baselines on the datasets currently available in the project. The two instantiated project systems are the Base and Instruct backbones after SFT followed by GRPO.

We evaluate the final systems on four datasets and emphasize two main metrics: ROUGE-2 for content overlap and absolute summary length distance for control accuracy. VietNews and WikiLingua provide the cleanest held-out single-document evidence. ViMs and VLSP remain informative for multi-document behavior, but their current test sets overlap with project training or validation pools and therefore appear in the paper with explicit exploratory caveats.

The paper makes three empirical contributions. First, it documents a lightweight Vietnamese controllable summarization pipeline from prompt construction through LoRA-SFT, GRPO alignment, and evaluation. Second, it reports a four-dataset benchmark that situates the final systems alongside historical baselines such as VietAI, GPT-4o, GPT-3.5-Turbo, and Qwen3-14B while keeping protocol caveats visible. Third, it includes focused analyses of the training stages and of the failure-prone behavior observed under earlier GRPO settings. The appendix preserves the supporting evidence behind these claims: the full within-project

stage tables, the historical configuration-bundle comparison, the exploratory multi-document baseline comparison, the sentence-control results, and the sentence-reward ablation.

2 Related Work

Vietnamese summarization benchmarks.

Vietnamese summarization has moved from benchmark construction toward stronger modeling, but the literature is still comparatively small. VNDS established an early Vietnamese news summarization benchmark, ViMs expanded the area with a high-quality multi-document abstractive corpus, WikiLingua added a multilingual how-to benchmark that includes Vietnamese, and the VLSP AbMuSu shared task provided another multi-document news setting for evaluation (Nguyen et al., 2019; Tran et al., 2020; Ladhak et al., 2020; Tran et al., 2023). Modeling papers then showed that pretrained Vietnamese seq2seq backbones matter: ViT5 improved Vietnamese abstractive summarization, and VieSum/BRIO reported further gains from BRIO-style training on a Vietnamese summarization corpus (Phan et al., 2022; Lam et al., 2023). In this line of work, however, the main objective is still generic summary quality, not explicit control over length or sentence budget.

Controllable and length-aware summarization.

Controllable summarization has progressed from architectural length conditioning to prompt-based instruction following. Early work controlled output length through decoder modifications or positional encodings, and later work generalized control to keywords or natural-language prompts (Kikuchi et al., 2016; Fan et al., 2018; Takase and Okazaki, 2019; He et al., 2020; Chan et al., 2021). More recent systems tighten the length problem further by targeting exact token or sentence counts and by benchmarking instruction-controllable summarization directly (Miculicich et al., 2023; Liu et al., 2023). Prompt-based multi-control methods and instruction-based length control show that LLMs can follow structured control text, but they also show that precise adherence is still fragile, especially outside English or under multiple simultaneous constraints (Jie et al., 2024; Juseon-Do et al., 2024). This project sits in that gap: it studies reference-conditioned Vietnamese control with coupled length and sentence targets rather than free-form user preference prompting.

Post-training for summarization behavior.

Our training recipe belongs to the broader post-training line rather than to architecture changes. LoRA makes adaptation practical by freezing the backbone and learning low-rank adapters, which is attractive when the desired behavior is narrow and the model is large (Hu et al., 2021). Summarization work with learned rewards has also shown that optimizing a better reward signal can outperform ROUGE-only training, as in learning from human feedback (Stiennon et al., 2020; Böhm et al., 2019). Recent policy-optimization methods such as GRPO further reduce the memory cost of on-policy alignment and have become common in open-model post-training; in that sense, our custom rollout-reward loop is best viewed as a lightweight adaptation of this broader family rather than a new optimizer (Shao et al., 2024). The contribution here is the application of these ideas to controllable Vietnamese summarization, not a new post-training algorithm.

Reward design and reward hacking. Once a model is optimized against a proxy reward, the reward itself becomes a design object. The AI safety literature uses the terms reward hacking and specification gaming for cases where optimizing the proxy produces superficially successful but substantively bad behavior (Amodei et al., 2016; Skalse et al., 2022). In summarization, reward-model work has long shown that ROUGE is an imperfect proxy for human judgment, and later analyses of reward-model overoptimization make the same point more sharply: increasing proxy reward can stop correlating with true quality (Gao et al., 2022). This literature motivates the content gate, degenerate-output filter, and multiplicative reward structure used in our GRPO setup, and it also explains why we treat the changes between the pre-safeguard and safeguarded configurations as a package-level mitigation rather than a single-component ablation.

3 Datasets

3.1 Task and Prompt Setting

Each example consists of a source document or document cluster x , a control instruction c , and a reference summary y^* . The control instruction specifies a target length and sentence budget derived from the reference summary statistics, so the task is best viewed as *reference-conditioned control* rather than arbitrary user-defined control. At

Dataset	Task	Test	Role	Caveat
VietNews	Single-document news	2,000	Main	Held-out single-doc. evidence; small duplicate caveat
WikiLingua	Single-document how-to	500	Main	Held-out single-doc. evidence
ViMs [†]	Multi-document news clusters	300	Explor.	Test overlaps 240 train and 60 val. clusters
VLSP [†]	Multi-document summarization	300	Explor.	Test overlaps 285 train and 15 val. items

Table 1: Datasets used in the report. VietNews and WikiLingua provide the strongest held-out evidence. ViMs and VLSP remain visible in the benchmark tables but are exploratory because their current test pools overlap with project training or validation data.

inference time, the model receives the source and the control instruction and produces $\hat{y}$.

The instruction schema is shared across SFT, GRPO, and evaluation. All prompts are built by the project’s data augmentation pipeline. Word counts are computed as whitespace-delimited tokens, and sentence counts are computed from end punctuation. SFT always uses the approximate constraint template family, GRPO rotates across approximate, range, and upper-bound templates to diversify the control language, and evaluation fixes the approximate template family for deterministic testing against the frozen artefacts.

3.2 Dataset Inventory and Validity Boundary

Table 1 summarizes the four datasets used in the report. VietNews and WikiLingua are the only held-out single-document datasets and therefore support the main generalization claims. ViMs and VLSP remain in the main benchmark because they reveal how the final systems behave on multi-document inputs, but they are explicitly marked exploratory throughout the paper.

After preprocessing, SFT uses 111,150 samples: 96,911 VietNews items whose titles contain at least 10 whitespace-delimited words, 13,999 WikiLingua items, and 240 ViMs clusters. GRPO uses a broader 119,942-sample pool: all 105,418 VietNews training items, the same 13,999 WikiLingua items, 285 VLSP items, and the same 240 ViMs clusters. Appendix A.5 records the exact preprocessing rules and Appendix A.6 shows compact prompt examples.

4 Method

4.1 Controllable Prompt Generation

The project constructs chat-style prompts rather than plain source-target pairs. For SFT, each training example contains system, user, and assistant turns. For GRPO and evaluation, the assistant response is removed and the model must generate a summary that respects the requested control budget. The three template families are approximate constraints (for example, “around X words”), explicit ranges, and upper bounds. This prompt diversification is one of the practical differences between the SFT stage and the GRPO stage.

4.2 Supervised Fine-Tuning

The SFT stage converts each processed $\{source, target\}$ pair into an instruction-following chat example. All canonical Qwen3 runs use LoRA with rank 32, $lora_alpha = 32$, and dropout 0.05 over the attention and MLP projection layers. The main SFT runs use a 3,072-token context length, learning rate 5×10^{-5} , warmup ratio 0.1, two epochs, and $disable_thinking = True$. In practice, this stage is responsible for the largest single improvement in instruction following and summarization quality in the project.

4.3 GRPO Alignment

GRPO is implemented as a custom rollout-and-reward loop rather than through a stock trainer abstraction. Each prompt is decoded multiple times, scored by the reward model, and updated against a frozen reference model derived from the same backbone. The project studies two initialization modes. *Fresh* GRPO attaches a new LoRA adapter directly to the pretrained backbone, while *SFT-initialized* GRPO warm-starts from the SFT checkpoint. In the final paper narrative, fresh GRPO is treated as an ablation baseline and SFT-initialized GRPO is treated as the intended deployment recipe.

Within each update, the trainer formats a batch of prompts, samples K completions per prompt, computes rewards $r_{i,k}$, and normalizes them within each prompt group:

$$A_{i,k} = \frac{r_{i,k} - \bar{r}_i}{\max(\sigma_i, 0.01)}, \quad (1)$$

where i indexes prompts and k indexes the sampled completions for prompt i . The policy is then updated with a PPO-style clipped objective over

generated tokens plus a KL penalty to the frozen reference model:

$$L_{\text{policy}} = -\frac{1}{|M|} \sum_{i,k,t} \min\left(\rho_{i,k,t} A_{i,k}, \text{clip}(\rho_{i,k,t}, 1 - \epsilon, 1 + \epsilon) A_{i,k}\right) m_{i,k,t}, \quad (2)$$

$$L = L_{\text{policy}} + \beta L_{\text{KL}}, \quad (3)$$

where $\rho_{i,k,t} = \exp(\log \pi_{\theta} - \log \pi_{\text{old}})$ and $m_{i,k,t}$ masks generated tokens up to the first EOS token. In the implementation, π_{old} is the detached current policy within the same on-policy step, and the KL term is computed token-wise against the frozen reference model. This makes the method a lightweight GRPO-style rollout-and-update loop tailored to the present summarization setting rather than a wrapper around a stock trainer.

4.4 Reward Design and Anti-Hacking Safeguards

The reward combines content quality with explicit control adherence. Content quality is measured as

$$R_{\text{acc}} = 0.5 \text{ROUGE-1}_{F1} + 0.5 \text{ROUGE-L}_{F1}. \quad (4)$$

Length and sentence rewards, denoted R_{len} and R_{sent} , are computed from the instruction-specific tolerance rules described in Section 3.1. The final reward uses a multiplicative gate:

$$R_{\text{total}} = R_{\text{acc}} \times (1 + 0.3R_{\text{len}} + 0.2R_{\text{sent}}), \quad (5)$$

which makes content quality a hard prerequisite: when $R_{\text{acc}} = 0$, the total reward is zero regardless of constraint satisfaction.

Because constraint-based rewards are susceptible to reward hacking where a model learns to satisfy length and sentence budgets with garbage content the reward function is embedded in a defense-in-depth strategy with three layers. At the **generation** layer, *repetition_penalty* is kept at 1.0 and *no_repeat_ngram_size* at 0. Preliminary experiments showed that higher values of either parameter, when applied by the HuggingFace decoder over the full $\approx 2,000$ -token input sequence, penalized the model for reusing source-article vocabulary and pushed decoding toward out-of-distribution symbols, which is especially damaging for short Vietnamese summaries where

named entities and key terms must appear verbatim. At the **detection** layer, a post-hoc filter `_is_degenerate()` examines each rollout with five sequential checks: empty or whitespace-only output; single-token blobs longer than 25 characters without spaces; fewer than three alphabetic characters; alpha character ratio below 12% when exceeding 15 characters; and token repetition where one token accounts for over 60% of all tokens or unique tokens fall below 30%. Any output flagged by this filter receives $R_{\text{len}} = 0$ and $R_{\text{sent}} = 0$, so degenerate rollouts cannot earn any constraint bonus; under the multiplicative gate, only residual content overlap captured by R_{acc} can remain. At the **formula** layer, the multiplicative gate (Equation 2) structurally prevents constraint-only optimization: no amount of length or sentence adherence can compensate for zero content quality. Additionally, the upper-bound constraint “khong qua X tu” enforces a floor of $\max(3, 0.15 \times \text{max_wc})$ words; outputs shorter than this floor receive $R_{\text{len}} = 0$, closing the loophole of satisfying the constraint with one- or two-word garbage.

4.5 GRPO Variant Nomenclature and Analysis Setups

The experiments vary two independent aspects: initialization path and GRPO configuration bundle. *Fresh GRPO* applies GRPO directly to a pretrained backbone without an SFT stage, whereas *SFT-initialized GRPO* warm-starts GRPO from the corresponding SFT checkpoint. Table 2 defines the compact configuration labels used from this point onward.

Label	Defining properties	Paper role
v3 (pre-safeguard)	Linear reward; <i>repetition_penalty</i> = 1.3; <i>no_repeat_ngram_size</i> = 3	Reward-hacking failure baseline
v4 (historical bundle)	$K = 4$, learning rate 5×10^{-7} , $\beta = 0.15$	Bundle comparison and sentence- reward ablation
v5 (canonical bundle)	$K = 8$, learning rate 2×10^{-6} , $\beta = 0.04$; gated reward and safeguards	Main paper configuration

Table 2: Compact labels for the GRPO configuration bundles used in the paper. Each label denotes a bundle of settings and, where applicable, system-level safeguards; the labels should not be interpreted as variants differing in only one hyperparameter.

The canonical v5 runs use 800 training steps and the same LoRA adapter shape as SFT. The two final project systems differ only in their underlying Qwen3-4B backbone family: Base or Instruct. Earlier GRPO settings are retained only as historical analysis points. Appendix A.6 records the exact prompt templates and launch-script presets used to instantiate the canonical runs, while Appendix A.2 preserves the historical v4-v5 bundle comparison.

5 Experimental Results

5.1 Experimental Protocol

All tables in this paper are filled from frozen project artefacts rather than reruns. The main project evaluation, the v3 comparison, and the no-sentence ablation are drawn from three frozen evaluation snapshots produced at different stages of the project. Evaluation uses *temperature* = 0.3, *top_p* = 0.9, *do_sample* = True, and *max_new_tokens* = 256; no evaluation seed was saved. The main paper reports only ROUGE-2 and absolute summary length distance. Sentence-control results, relative length error, dataset-processing details, and prompt examples are moved to the appendix, where Appendix A.4 keeps the sentence-control table, Appendix A.7 records the no-sentence ablation, and Appendix A.1 preserves the full within-project variant tables.

The external baseline rows in Tables 3 and 4 come from a historical benchmark collection maintained alongside the project. They are use-

ful contextual comparisons, but they are not protocol-matched reruns under the frozen evaluation pipeline used by the project systems. Accordingly, VietNews and WikiLingua remain the strongest evidence for model comparison inside the present report, while ViMs and VLSP are read as exploratory diagnostics; Appendix A.3 retains the full baseline-relative comparison for those datasets.

5.2 Main Benchmark Against External Baselines

Tables 3 and 4 provide the headline benchmark of the paper. The two project rows are the final SFT + GRPO v5 systems, and the remaining rows are comparison baselines for Section 5.2.

Figure 1 summarizes the same benchmark in terms of percentage improvement of the two final systems over each baseline. Each dataset appears explicitly as a two-column block ($\Delta R2$ and LenRed), so the figure complements Tables 3 and 4 without replacing their raw per-dataset values.

The two main tables show the raw benchmark values, while Figure 1 makes the comparison direction explicit. On VietNews and WikiLingua, both final systems improve strongly over the raw Qwen3 baselines and compare favorably to several external rows with available length-control values; VietAI remains the strongest ROUGE-2 row on both datasets, and the external baselines should still be read as contextual because they are not protocol-matched reruns. The same figure also shows that the method is not merely beating unrelated baselines: relative to the raw Qwen3 backbones, both final systems make especially large ROUGE-2 gains on WikiLingua and strong length-distance reductions on both single-document datasets.

ViMs and VLSP remain visible rather than hidden. Figure 1 shows that both final systems still improve over the raw Qwen3 baselines on ViMs and VLSP ROUGE-2, but they remain substantially below the strongest external multi-document baselines such as GPT-4o and Qwen3-14B on that metric. The ViMs/VLSP length-control picture is mixed rather than uniformly positive, especially for the Instruct branch against stronger baselines. Appendix Table 11 retains the full baseline-relative percentage comparison for readers who want the detailed ViMs/VLSP view.

Between the two backbone instantiations, the In-

Model	VietNews	WikiLingua	ViMs [†]	VLSP [†]
<i>Project systems</i>				
Qwen3-4B-Base + SFT + GRPO v5	28.97	<u>26.35</u>	18.16	17.46
Qwen3-4B-Instruct + SFT + GRPO v5	<u>29.62</u>	<u>25.59</u>	21.27	24.39
<i>Raw Qwen3 baselines (frozen project protocol)</i>				
Qwen3-4B-Base (raw)	12.13	3.18	13.46	9.49
Qwen3-4B-Instruct (raw)	16.48	7.44	17.83	15.51
<i>Historical external baselines</i>				
VietAI	34.24	33.12	–	–
GPT-4o	21.61	20.65	<u>44.26</u>	<u>43.37</u>
GPT-3.5-Turbo	28.13	21.09	19.28	35.79
Qwen3-14B	18.51	20.24	44.38	44.27
Llama3.3-70B-Instruct	22.17	19.40	37.54	39.02
Phi4-14B	13.17	14.28	41.85	42.31
Sailor-20B-chat	17.70	18.74	39.47	40.96
VinBigdata (7B)	20.59	21.02	37.98	40.23

Table 3: ROUGE-2 (%) across the four report datasets. Bold marks the best reported value in each column and underlining marks the second-best reported value across both project and baseline rows. The two final systems and the raw Qwen3 baseline rows are evaluated under the frozen project protocol; the remaining external rows are drawn from a historical benchmark collection and should be read as contextual rather than protocol-matched reruns. ViMs and VLSP remain exploratory.

struct branch is the stronger content model overall, winning ROUGE-2 on VietNews, ViMs, and VLSP. The Base branch is slightly stronger on WikiLingua ROUGE-2 and also yields the best single-document length distance, especially on WikiLingua. This is why the paper keeps both final rows in the main benchmark rather than collapsing the method to one backbone only.

5.3 Within-project Stage-wise Ablation

Table 5 compresses the internal project comparison into one stage-wise ablation over the two held-out single-document datasets. Its purpose is to analyze the observed effect of the two-stage post-training pipeline (SFT + GRPO v5) relative to SFT-only, pretrained-only, and fresh-GRPO variants. Appendix A.1 provides the full four-dataset within-project tables, while Appendix A.2 records the historical v4-v5 bundle comparison that motivated the final GRPO choice. To better visualize this multi-stage interaction on the primary dual objectives retaining content quality while enforcing a strict length budget we plot the joint ROUGE-2 and absolute summary length distance trajectory in Figure 2.

The stage-wise comparison yields three observations about the evaluated training recipe:

SFT provides the largest stage-wise gain. Both pretrained backbones have substantially lower ROUGE-2 than their SFT variants on the two held-out datasets. The largest observed gains occur after SFT, making it the dominant improvement stage in the present comparison. The subsequent GRPO results are therefore interpreted relative to this strong SFT baseline. GRPO v5 then acts as an *alignment refiner* in the evaluated systems. It improves ROUGE-2 for both SFT-initialized backbones. For the Base backbone it also reduces length distance on both held-out datasets; for the Instruct backbone, length distance remains close to SFT-only and is slightly worse. For instance, on the Base backbone, GRPO refinement reduces length deviation on VietNews by 80% (from 5.73 to 1.16 words), while ROUGE-2 rises from 26.61 to 28.97.

Fresh GRPO underperforms SFT initialization. Applying GRPO directly to the pretrained checkpoint (Fresh GRPO v5, without the intermediate SFT step) underperforms SFT-initialized GRPO on both held-out datasets. As shown by the isolated markers in Figure 2, the Fresh GRPO models reach only 20.73 and 18.67 ROUGE-2 on VietNews, compared with 28.97 and 29.62 after SFT initialization. This result establishes the em-

Model	VietNews	WikiLingua	ViMs [†]	VLSP [†]
<i>Project systems</i>				
Qwen3-4B-Base + SFT + GRPO v5	1.16	8.35	89.23	100.50
Qwen3-4B-Instruct + SFT + GRPO v5	<u>1.52</u>	<u>10.25</u>	84.45	82.71
<i>Raw Qwen3 baselines (frozen project protocol)</i>				
Qwen3-4B-Base (raw)	33.03	61.59	92.01	120.33
Qwen3-4B-Instruct (raw)	4.01	23.42	<u>59.80</u>	70.17
<i>Historical external baselines</i>				
VietAI	–	–	–	–
GPT-4o	8	11	187	58
GPT-3.5-Turbo	18	17	47	30
Qwen3-14B	50	45	131	<u>34</u>
Llama3.3-70B-Instruct	14	13	186	73
Phi4-14B	205	214	166	134
Sailor-20B-chat	38	43	268	41
VinBigdata (7B)	115	89	98	82

Table 4: Absolute summary length distance across the four report datasets. Lower is better. Bold marks the best reported value in each column and underlining marks the second-best reported value across both project and baseline rows. The two final systems and the raw Qwen3 baseline rows use the frozen project protocol; the remaining external rows are contextual historical comparisons only. ViMs and VLSP remain exploratory.

Model	VietNews ROUGE-2	WikiLingua ROUGE-2	VietNews LenDist	WikiLingua LenDist
Base pretrained	12.13	3.18	33.03	61.59
Base + SFT	26.61	24.43	5.73	14.18
Base + fresh GRPO v5	20.73	9.71	3.80	22.03
Base + SFT + GRPO v5	<u>28.97</u>	26.35	<u>1.16</u>	8.35
Instruct pretrained	16.48	7.44	4.01	23.42
Instruct + SFT	26.70	23.93	1.09	<u>9.87</u>
Instruct + fresh GRPO v5	18.67	8.09	2.95	16.12
Instruct + SFT + GRPO v5	29.62	<u>25.59</u>	1.52	10.25

Table 5: Compact within-project ablation on the two held-out single-document datasets. This table is included to explain the chosen training recipe, not as the paper’s headline leaderboard.

pirical importance of SFT warm-starting in the present setup. The multiplicative quality gate is a plausible contributing mechanism, but the current experiments do not isolate it causally.

Observed trade-offs and backbone choice.

The trajectory curves in Figure 2 visualize the two-stage warm-start as a clear shift towards the top-right corner. For the Base branch, the absolute length distance decreases by 89.9% on the macro average (47.31 words to 4.76 words) coupled with a +261.3% relative gain in macro ROUGE-2. For the Instruct branch, macro length distance is slashed by 57.1% (13.72 to 5.89 words) alongside a +130.8% relative macro ROUGE-2 gain. Among the evaluated systems, the Instruct-initialized pipeline has the highest VietNews ROUGE-2 (29.62), whereas the Base-initialized

pipeline has the lowest single-dataset length distance (1.16 words). Together, these results show consistent gains from the two-stage recipe across the two backbone families evaluated here.

5.4 Reward-hacking Mitigation Analysis

Reward hacking occurs when a model optimizes the reward signal without improving output quality. In v3 (the pre-safeguard bundle), three interacting design flaws created conditions under which this behavior emerged. First, v3 used a **linear reward sum** $R_{\text{total}} = w_{\text{acc}}R_{\text{acc}} + w_{\text{len}}R_{\text{len}} + w_{\text{sent}}R_{\text{sent}}$, which allowed the constraint terms to contribute independently of content: a model could score ≈ 0.50 by satisfying length and sentence budgets with garbage while ignoring R_{acc} . Second, $\text{repetition_penalty} = 1.3$ was applied by the HuggingFace decoder over the full $\approx 2,000$ -

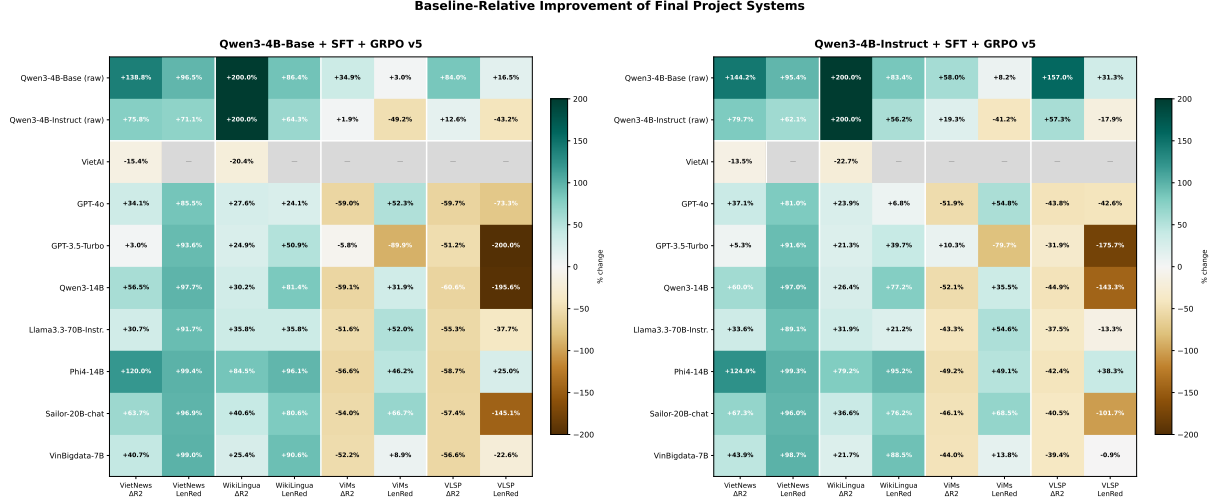


Figure 1: Baseline-relative improvement summary for the two final project systems. Each dataset is shown explicitly as a two-column block: positive ΔR_2 means higher ROUGE-2, positive LenRed means smaller absolute summary length distance, and gray cells mark unavailable values. The first two rows are raw Qwen3-4B baselines, followed by external baselines. Cell text shows the exact percentage change, while color intensity is clipped for readability.

token input sequence, penalizing the model for reusing any vocabulary from the source article and pushing generation toward out-of-distribution symbols. Third, *no_repeat_ngram_size* = 3 blocked source trigrams, forbidding named entities and key Vietnamese content words. The resulting failure modes, concentrated in the Base branches where the pretrained backbone lacks the instruction-following prior of the Instruct model, included repetition loops (e.g., *assistant* repeated 128 times, $ROUGE-2 = 0.0$), punctuation padding (245 words of periods, $LenErr = 1189\%$), digit blobs that satisfied “*khong qua X tu*” with $R_{len} = 1.0$ and $R_{acc} = 0$, and comma-separated token loops carrying no semantic content. Figure 3 illustrates the two most severe modes qualitatively. Appendix A.2 records the historical bundle-level v4-v5 comparison, and Appendix A.7 keeps the sentence-reward ablation that remains inconclusive under the frozen sampling protocol.

The canonical v5 bundle addresses each root cause through the defense-in-depth safeguards described in Section 4.4: the generation-layer fixes restore natural decoding, the detection-layer filter catches degenerate outputs, and the multiplicative gate structurally prevents constraint-only optimization.

Table 6 reports the quantitative evidence. Base fresh shows the most dramatic recovery: degenerate rate collapses from 23.81% to 0.32% ($\approx 74\times$ reduction), $R_{acc} = 0$ drops from 27.16% to 1.94%,

and ROUGE-2 rises from 10.58 to 18.66. Base SFT-init suffered severe length explosion under v3 ($LenErr = 425\%$), which v5 cuts to 14.94% while raising ROUGE-2 from 18.03 to 26.38. The Instruct branches were substantially more robust under v3—degenerate rates remained below 1%—owing to their chat-tuning prior; their v3-to-v5 gains are correspondingly incremental, with Instruct SFT-init v5 reaching the highest mean R_{acc} (0.4213) and ROUGE-2 (27.65). Across all eight configurations, the v5 safeguards eliminate catastrophic failure modes without harming any branch, and the recovery magnitude scales inversely with v3 baseline quality.

These results establish v5 as the stable canonical GRPO configuration. The Instruct branches’ near-immunity to degeneration reinforces the finding from Section 5.3 that an SFT warm-start or equivalent generative prior is a prerequisite for effective GRPO in this setting. The v3-to-v5 improvement is properly understood as the effect of a **mitigation package**—generation, detection, and formula-level safeguards acting jointly—rather than an ablation of any single component. Appendix A.4 retains the sentence-control results that justify keeping sentence reward as an explicit metric, and Appendix A.7 shows that removing that term does not support a clean causal claim under the current evaluation protocol.

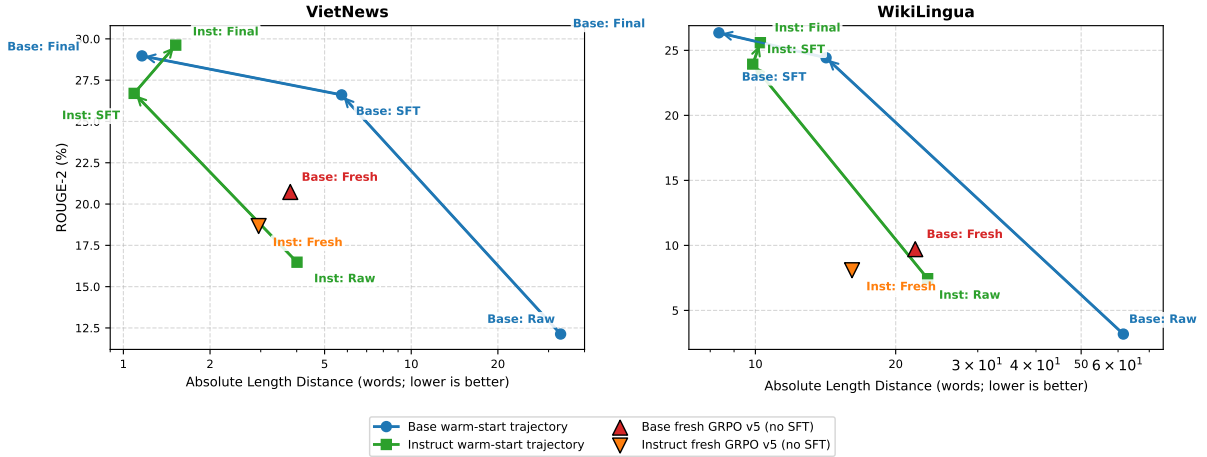


Figure 2: Two-panel trade-off trajectory showing ROUGE-2 versus absolute summary length distance for the Base and Instruct backbones on VietNews (left) and WikiLingua (right). Arrowed lines display the observed stage sequence (Raw $\rightarrow$ SFT $\rightarrow$ SFT+GRPO v5), while standalone markers show the lower content scores of Fresh GRPO v5 without SFT initialization.

Run	ROUGE-2	LenErr%	Deg.%	$R_{acc} = 0\%$	Mean R_{acc}
Base fresh v3	10.58	163.16	23.81	27.16	0.1854
Base fresh v5	18.66	29.05	0.32	1.94	0.3160
Base SFT-init v3	18.03	425.00	14.77	1.16	0.2508
Base SFT-init v5	<u>26.38</u>	14.94	2.71	0.45	<u>0.4009</u>
Instruct fresh v3	15.00	30.08	0.03	1.48	0.2768
Instruct fresh v5	16.69	22.78	<u>0.03</u>	1.58	0.2979
Instruct SFT-init v3	25.06	19.94	0.94	1.19	0.3837
Instruct SFT-init v5	27.65	<u>15.85</u>	1.42	<u>0.45</u>	0.4213

Table 6: Reward-hacking mitigation from v3 to v5, computed from stored generations with the current reward implementation. Deg.% is the fraction flagged by `_is_degenerate()`; $R_{acc} = 0\%$ is the fraction receiving zero content reward; Mean R_{acc} is the average content reward. The comparison is a package-level ablation and does not isolate any single change.

6 Conclusion

This paper reframes the project as a method-first study of lightweight post-training for controllable Vietnamese summarization. The central result is that a two-stage recipe, combining instruction-conditioned LoRA-SFT with custom GRPO v5, improves single-document performance within the evaluated Qwen3-4B systems on both ROUGE-2 and absolute summary length distance. In the main benchmark, the final project systems are competitive with, and often above, several historical comparison rows on VietNews and WikiLingua while maintaining low length distance; those rows remain contextual rather than protocol-matched comparisons.

The within-project comparisons characterize the contributions of the evaluated training stages.

SFT is the dominant improvement stage, fresh GRPO is not a substitute for supervised instruction tuning, and the v3-to-v5 mitigation package sharply reduces failure-prone behavior in the branches that were previously unstable.

At the same time, the paper keeps its claim boundary explicit. VietNews and WikiLingua are the only datasets that support strong held-out conclusions, ViMs and VLSP remain exploratory because of split overlap, and the external baseline rows are contextual rather than protocol-matched reruns. The main takeaway is therefore narrow: the present SFT + GRPO v5 pipeline shows clear within-project gains for single-document controllable Vietnamese summarization, while multi-document generalization remains future work. The appendix keeps the supporting material for readers who want the fuller

Case	Reference	v3 output (failure)	v5 output (recovered)
Base fresh idx 14	Điều tra vụ anh rể dùng dao đâm chết em vợ vào rạng sáng	Output <i>assistant assistant assistant assistant assistant assistant ... (×128 tokens)</i>	Output <i>Anh C. bị anh rể đâm chết tại nhà.</i>
		Failure signature • Pure repetition loop — no content • ROUGE-2 = 0.00 • LenErr = 814.29% • R_acc = 0.00	Recovered properties • Short, content-bearing headline • Named entity retained • Budget restored • ROUGE-2 = 0.19
Base SFT-init idx 7	5 năm chưa xử xong vụ án “lạ” có hơn 30 luật sư bào chữa miễn phí	Output <i>Bị cáo Tuyết bị cấp sơ thẩm 2 lần tuyên phạt 12 năm tù . Bị cáo Tuyết kháng cáo , nên 2 lần vụ án được đưa ra xét xử phúc thẩm và đều có chung 1 kết quả , huỷ án</i>	Output <i>Vụ án “lạ” : Huỷ án 2 lần , bị cáo vẫn bị tạm giam 5 năm !</i>
		Failure signature • 245 words — punctuation-heavy blow-up • Meaningful start → punctuation flood • LenErr = 1189.47% • ROUGE-2 = 0.034	Recovered properties • Compact, accurate rewrite • Budget restored • Content preserved • ROUGE-2 = 0.22

Figure 3: Paired v3–v5 examples of the two most severe reward-hacking modes from VietNews. Each card shows a reference summary (gray), the failed v3 output (red), and the recovered v5 output (green), together with failure or recovery indicators. Left: repetition loop (Base fresh; *assistant* repeated 128 times). Right: punctuation-driven length explosion (Base SFT-init; 245 words of periods).

evidence trail: Appendix A.1, Appendix A.3, Appendix A.4, and Appendix A.7.

A Appendix

A.1 Full Within-project Variant Tables

Tables 7–9 provide the full four-dataset view for all canonical Qwen3 project variants. These tables are retained as internal context, but they are not the headline benchmark of the paper.

Model	VietNews	WikiLingua	ViMs [†]	VLSP [†]
Base pretrained	12.13	3.18	13.46	9.49
Base + SFT	26.61	24.43	25.65	28.14
Base + fresh GRPO v5	20.73	9.71	20.28	18.14
Base + SFT + GRPO v5	<u>28.97</u>	26.35	18.16	17.46
Instruct pretrained	16.48	7.44	17.83	15.51
Instruct + SFT	26.70	23.93	<u>24.71</u>	<u>27.03</u>
Instruct + fresh GRPO v5	18.67	8.09	18.43	16.05
Instruct + SFT + GRPO v5	29.62	<u>25.59</u>	21.27	24.39

Table 7: ROUGE-2 (%) across all four datasets for the canonical Qwen3 project variants.

Model	VietNews	WikiLingua	ViMs [†]	VLSP [†]
Base pretrained	33.03	61.59	92.01	120.33
Base + SFT	5.73	14.18	66.45	<u>61.64</u>
Base + fresh GRPO v5	3.80	22.03	71.77	88.02
Base + SFT + GRPO v5	<u>1.16</u>	8.35	89.23	100.50
Instruct pretrained	4.01	23.42	59.80	70.17
Instruct + SFT	1.09	<u>9.87</u>	64.55	62.22
Instruct + fresh GRPO v5	2.95	16.12	<u>63.05</u>	72.91
Instruct + SFT + GRPO v5	1.52	10.25	84.45	82.71

Table 8: Absolute summary length distance across all four datasets for the canonical Qwen3 project variants.

Model	VietNews	WikiLingua	ViMs [†]	VLSP [†]
Base pretrained	205.89	174.67	42.61	47.93
Base + SFT	31.19	24.72	38.69	27.53
Base + fresh GRPO v5	22.36	50.94	31.89	34.33
Base + SFT + GRPO v5	<u>6.43</u>	14.69	44.42	42.55
Instruct pretrained	24.50	57.47	26.07	<u>27.09</u>
Instruct + SFT	6.26	18.34	31.55	24.51
Instruct + fresh GRPO v5	17.08	38.67	<u>28.37</u>	28.76
Instruct + SFT + GRPO v5	9.24	<u>18.11</u>	39.14	32.85

Table 9: Relative length error (%) across all four datasets for the canonical Qwen3 project variants.

A.2 Historical Comparison of Two GRPO Configuration Bundles

This appendix records the two GRPO configuration bundles available in the stored artefacts: v4 ($K = 4$, learning rate 5×10^{-7} , $\beta = 0.15$) and v5 ($K = 8$, learning rate 2×10^{-6} , $\beta = 0.04$). Because only two operating points were evaluated, three hyperparameters change jointly, and the sampling-based evaluation has no saved seed, this comparison is not a controlled hyperparameter ablation or a systematic search.

Branch	v4 ROUGE-2	v5 ROUGE-2	v4 LenErr%	v5 LenErr%	v4 LenDist	v5 LenDist
Base fresh	10.93	15.22	91.67	36.65	26.70	12.90
Base SFT-init	26.25	27.66	17.88	10.56	7.05	4.80
Instruct fresh	12.39	13.38	33.98	27.88	11.65	9.50
Instruct SFT-init	26.23	27.60	13.09	13.68	6.00	5.85

Table 10: Historical comparison of two explored GRPO configuration bundles on VietNews and WikiLingua macro averages. These observations do not identify the effect of any individual hyperparameter or establish a globally optimal configuration.

Across the stored runs, v5 has higher ROUGE-2 in all four branches and lower length distance in all four branches. Relative length error is lower under v5 in three branches and slightly higher for Instruct SFT-init. These values are retained for transparency and hypothesis generation only, not as evidence for a causal configuration claim.

A.3 Exploratory Multi-document Baseline Comparison

Table 11 keeps the full baseline-relative comparison on ViMs and VLSP. It is retained in the appendix because it is useful diagnostic context, but it should not dominate the main method-first narrative of the paper.

Baseline	Qwen3-4B-Base + SFT + GRPO v5		Qwen3-4B-Instruct + SFT + GRPO v5	
	Δ ROUGE-2 $\uparrow$	LenDist reduction $\uparrow$	Δ ROUGE-2 $\uparrow$	LenDist reduction $\uparrow$
VietAI	—	—	—	—
GPT-4o	-59.4%	+22.6%	-47.9%	+31.8%
GPT-3.5-Turbo	-35.3%	-146.4%	-17.1%	-117.1%
Qwen3-14B	-59.8%	-15.0%	-48.5%	-1.3%
Llama3.3-70B-Instruct	<u>-53.5%</u>	+26.7%	<u>-40.4%</u>	+35.5%
Phi4-14B	-57.7%	+36.8%	-45.7%	+44.3%
Sailor-20B-chat	-55.7%	+38.6%	-43.2%	+45.9%
VinBigdata (7B)	-54.5%	-5.4%	-41.6%	+7.1%

Table 11: Exploratory relative change against each contextual baseline on the ViMs and VLSP macro average. Positive values are better for both columns, but the table is diagnostic only because both datasets overlap with project training or validation pools.

A.4 Sentence-control Results

Sentence-control results are retained here because the main paper focuses on ROUGE-2 and absolute summary length distance.

Model	VN Exact%	VN Tol.%	VN MAE	WL Exact%	WL Tol.%	WL MAE
Base pretrained	73.95	89.55	0.550	15.4	32.4	2.910
Base + SFT	91.40	96.55	4.299	58.6	83.8	1.674
Base + fresh GRPO v5	91.80	99.30	0.089	34.2	72.8	1.226
Base + SFT + GRPO v5	93.85	98.75	0.316	65.6	<u>95.6</u>	0.470
Instruct pretrained	93.95	<u>99.40</u>	0.067	34.2	64.6	1.582
Instruct + SFT	<u>95.45</u>	98.90	0.058	88.0	97.0	0.204
Instruct + fresh GRPO v5	94.85	99.60	<u>0.056</u>	46.8	72.4	1.178
Instruct + SFT + GRPO v5	96.95	99.05	0.041	<u>78.8</u>	95.6	<u>0.340</u>

Table 12: Sentence-control metrics recomputed from stored generations. “Tol.” counts a prediction as correct when the sentence count is within ± 1 of the requested budget.

A.5 Dataset Processing Details

The final JSONL artefacts are produced by the project’s data processing and augmentation modules. The most important preprocessing rules are listed below.

- Shared preprocessing truncates sources to 8,000 characters and summaries to 1,500 characters, then replaces underscore-based segmentation markers with spaces.
- VietNews uses the first line of each file as the target headline, the sapo plus body as source text, and drops the last line as caption-like trailing content.
- WikiLingua flattens *src* and *tgt* sentence lists into single strings.
- ViMs reads each document in a cluster and concatenates the documents into a single source with explicit boundary markers.
- VLSP flattens each cluster in the same style, prefers the abstractive *summary* field, and falls back to label-based extraction only when a summary is unavailable.
- SFT and GRPO intentionally use different training pools: SFT filters short VietNews titles and excludes VLSP, while GRPO keeps all VietNews and includes VLSP.

A.6 Pipeline Configurations and Prompt Templates

This subsection summarizes the end-to-end pipeline configuration used by the paper. The source of truth is the project’s canonical configuration files, training scripts, evaluation pipeline, and audit documentation. Prompt examples are shown in an ASCII-friendly transliteration for toolchain compatibility, but they follow the same instruction format used throughout the pipeline.

Process	Prompt shape	Template policy	Stored outputs
SFT	system + user + assistant	Always uses the approximate control template family	<i>messages</i> plus <i>meta</i>
GRPO train/val	system + user	Rotates three length templates and three sentence templates across prompts	<i>prompt</i> , <i>reference</i> , and <i>meta</i>
Evaluation/test	system + user	Deterministic test prompt; <i>make_test_prompt()</i> fixes template index 0	<i>prompt</i> , <i>reference</i> , and <i>meta</i>

Table 13: Prompt-construction rules used by the pipeline.

Signal	Template families	Notes
Length	<i>khoang X tu; trong khoang lo-hi tu; khong qua X tu</i>	SFT uses only the first form; GRPO rotates all three; eval fixes the first form
Sentence	<i>khoang X cau; trong khoang lo-hi cau; khong qua X cau</i>	Same rotation policy as the length templates

Table 14: Control-template families generated by *PromptAugmenter*.

Data augmentation and prompt construction. All prompt-bearing samples preserve metadata for the reward functions and analysis logs. The core fields are *target_length*, *target_sentences*, *length_requirement*, and *sentence_requirement*; the pipeline also tracks source and target length statistics such as *source_length*.

Setting	Canonical paper configuration
Backbone family	Qwen3-4B-Base or Qwen3-4B
LoRA adapter	$r = 32$, $\alpha = 32$, dropout 0.05; target modules <i>q/k/v/o_proj</i> and <i>gate/up/down_proj</i>
Sequence length	3072 on H200/A100; 2048 on V100
Training	$lr = 5 \times 10^{-5}$, cosine scheduler, <i>warmup_ratio</i> = 0.1, <i>epochs</i> = 2.0
Thinking suppression	<i>disable_thinking</i> = True
Batch calibration	Enabled before training
Checkpoint behavior	Model-family-specific output directory; resume from latest checkpoint when present

Table 15: Canonical SFT configuration used by the paper.

GPU tier	Start batch	Grad. accum.	Effective batch
H200	10	2	20
A100	6	3	18
V100	2	4	8

Table 16: SFT per-GPU starting presets used in the canonical training runs. Auto-calibration can adjust these values at runtime.

Canonical SFT configuration. The appendix tables intentionally follow the launch-script presets rather than the raw library defaults, because the launch scripts are the source of truth for the canonical paper runs.

Setting	Canonical paper configuration
Initialization	SFT-initialized warm start
Core v5 hyperparameters	$K = 8$, $lr = 2 \times 10^{-6}$, $\beta = 0.04$, <i>total_steps</i> = 800
Rollout decoding	<i>temperature</i> = 0.7, <i>top_p</i> = 0.9, <i>do_sample</i> = True, <i>max_new_tokens</i> = 80, <i>max_seq_length</i> = 3072
Reward weights	Accuracy 0.5, length 0.3, sentence 0.2
Stability settings	<i>gradient_checkpointing</i> = True, <i>disable_thinking</i> = True, auto-calibration enabled
Repetition controls	<i>repetition_penalty</i> = 1.0, <i>no_repeat_ngram_size</i> = 0

Table 17: Canonical GRPO v5 configuration used in the main paper results.

GPU tier	Start batch	Grad. accum.	Effective prompts/update
H200	4	4	16
A100	2	8	16
V100	1	16	16

Table 18: GRPO per-GPU starting presets used in the canonical training runs. Auto-calibration can refine these values at runtime.

Canonical GRPO v5 configuration. For reference only, the main ablation bundle changes are v4: $K = 4$, $lr = 5 \times 10^{-7}$, $\beta = 0.15$; v3 and v4 remain ablation settings and should not be read as the canonical appendix configuration.

Setting	Frozen evaluation protocol
Decoding	<i>temperature</i> = 0.3, <i>top_p</i> = 0.9, <i>do_sample</i> = True, <i>max_new_tokens</i> = 256
Batching	<i>batch_size</i> = 8
Prompt formatting	<i>apply_chat_template(..., enable_thinking = False)</i> when supported
Prompt truncation	Tokenizer truncates prompts to $3072 - \text{max_new_tokens}$
Evaluation inputs	Per-dataset test artefacts plus the combined test split
Metrics in main paper	ROUGE-2 and Length Distance; sentence-control kept in appendix
Randomness	One-shot low-temperature sampling; no saved seed

Table 19: Frozen evaluation configuration used to generate the paper tables.

Evaluation configuration. SFT prompt example.

System: Ban là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: khoảng 22 từ
- Số câu: khoảng 1 câu

Văn bản: <source article>

Assistant: <gold summary>

GRPO rollout prompt example.

System: Ban là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: trong khoảng 18–26 từ
- Số câu: trong khoảng 1–2 câu

Văn bản: <source article>

reference = <gold summary>

meta.target_length = 22

meta.target_sentences = 1

meta.length_requirement = “trong khoảng 18–26 từ”

meta.sentence_requirement = “trong khoảng 1–2 câu”

Evaluation/test prompt example.

System: Ban là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: khoảng 22 từ
- Số câu: khoảng 1 câu

Văn bản: <source article>

reference = <gold summary>

meta.dataset = “vietnews”

GRPO training and evaluation share the same prompt shape: both keep only the system and user turns, store the gold summary separately as *reference*, and preserve the control metadata in *meta*. The differences are that GRPO rotates all template families during rollout training, whereas evaluation fixes the deterministic approximate template, and SFT is the only stage that includes an assistant turn inside the serialized chat sample.

A.7 No-sentence-reward Ablation

The no-sentence ablation is informative but not decisive. Both sides of the comparison still use sampling without a saved evaluation seed, so the correct conclusion is *inconclusive* rather than a clean causal claim about sentence reward.

Branch	w_{sent}	ROUGE-2	LenErr%	Sent. Exact%	Sent. MAE
Base fresh v4	0.2	13.88	76.44	59.42	1.136
Base fresh ablation	0.0	13.99	76.50	60.32	<u>1.163</u>
Base SFT-init v4	0.2	<u>26.71</u>	<u>20.82</u>	<u>73.39</u>	2.009
Base SFT-init ablation	0.0	26.85	17.21	73.65	1.452

Table 20: No-sentence-reward ablation computed from a frozen evaluation snapshot. The differences are too small under the current sampling-based protocol to support a clean causal statement.

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